



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

--	--	--	--

INDEX
NUMBER

--	--	--	--

CLASS

H1 BIOLOGY

8875/02

Paper 2

14 September 2011

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

Circle the question number in the table on the right.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 10
2	/ 12
3	/ 7
4	/ 11
5 or 6*	/ 20
Total	/ 60

This Question Paper consists of **13** printed pages.

Section A

Answer **all** the questions in this section.

1.

PART I

Fig. 1.1 is a high power electronmicrograph of a cell.

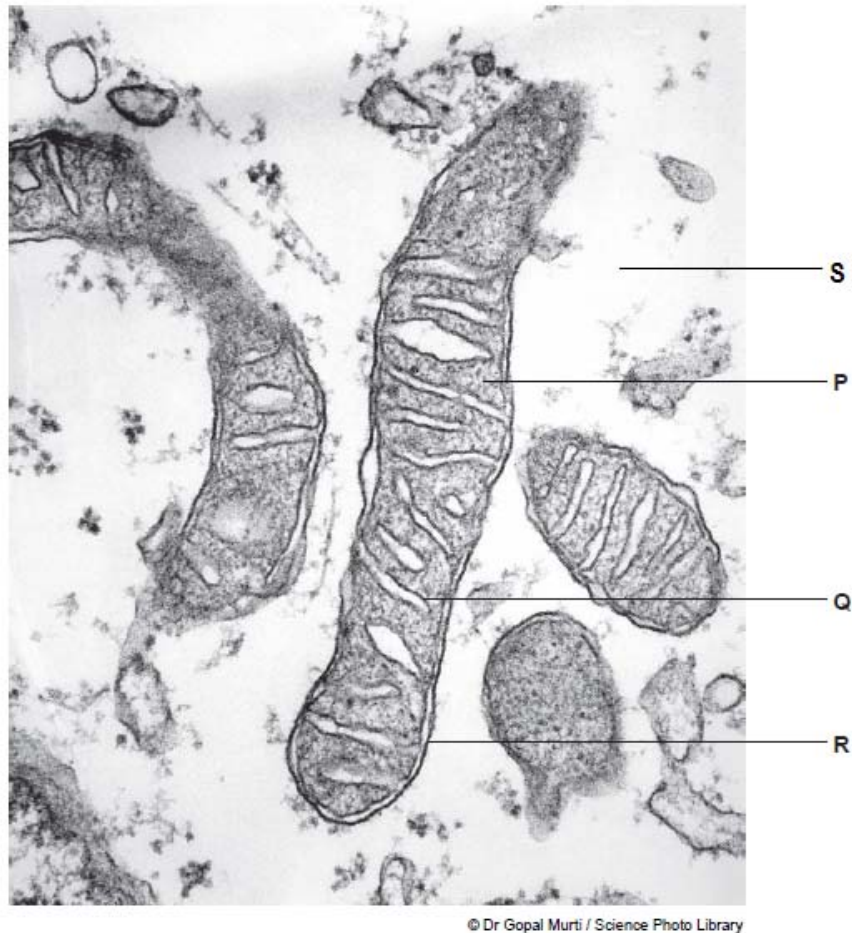


Fig. 1.1

- (a) With reference to **Fig. 1.1**, where in the cell does each of the following occur? Indicate your answer with the appropriate letter. [2]

(i) *Krebs cycle*

Oxidative phosphorylation

- (ii) There are differences in the ways in which ATP is made in glycolysis and in the electron transport chain.

Give **one** of these differences.

[2]

1.

PART II

Lipid vesicles (i.e. spherical lipid bilayers) that contain $\text{Na}^+\text{-K}^+$ pumps as the sole membrane protein were prepared. For the sake of simplicity, assume that each pump transports one Na^+ one way and one K^+ the other way in each pumping cycle as shown in **Fig. 1.2**. Each $\text{Na}^+\text{-K}^+$ pump is oriented so that the portion of the molecule that normally faces the cytosol faces the outside of the vesicle.

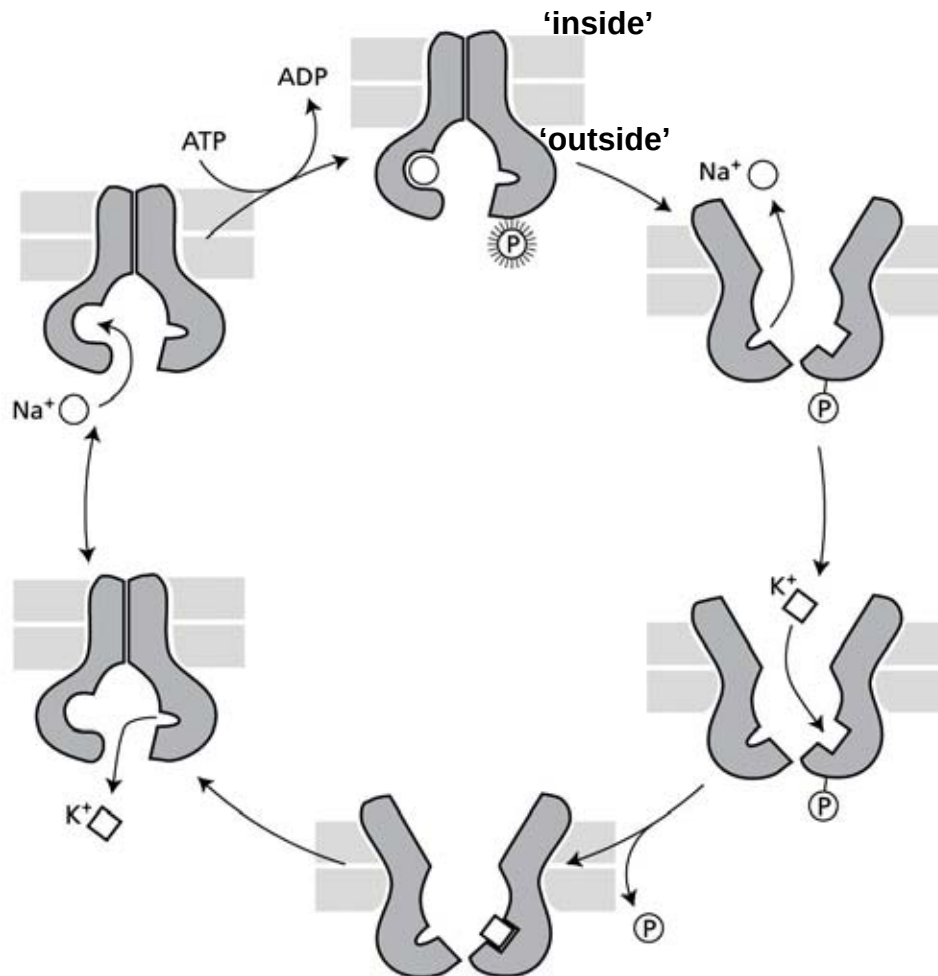


Fig. 1.2

With reference to **Fig. 1.2**, explain what would happen under each of the following conditions.

- (b) The solution inside and outside the vesicles contains both Na^+ and K^+ ions but no ATP.

[1]

1. (c) The solution inside the vesicles contains both Na^+ and K^+ ions; the solution outside contains both ions, as well as ATP. [2]

- (d) The solution inside contains Na^+ ; the solution outside contains Na^+ and ATP. [3]

[Total: 10]

2.

PART I

Fig. 2.1 is a diagrammatic representation of the process of transcription.

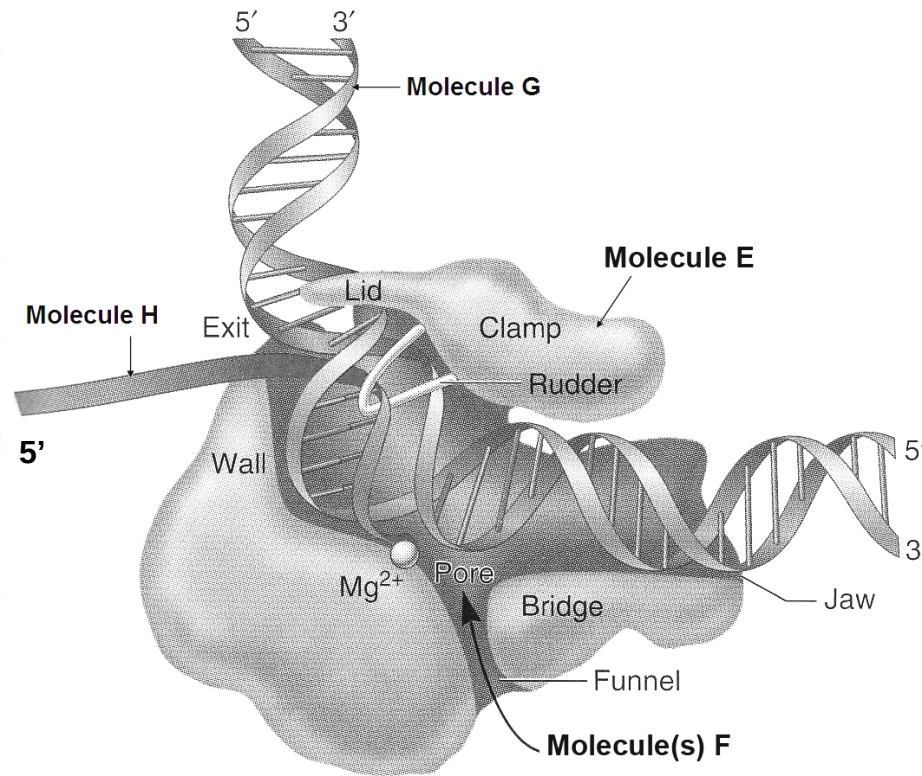


Fig. 2.1

- (a) Describe how the structure of molecule **E** is adapted to its role in transcription. [2]

- (b) Describe two ways by which the structure of molecule(s) **F**, which is / are adapted for transcription, differs from that of amino acids, which are adapted for translation. [2]

1.

2.

2. (c) Using evidence from **Fig. 2.1**, explain why the process of transcription is conservative. [2]

- (d) Explain why molecule **H** is shorter than molecule **G**. [2]

- (e) Molecule **E** is inhibited by the toxin, α -amanitin. **Fig. 2.2** shows the effect when α -amanitin attaches to this molecule.

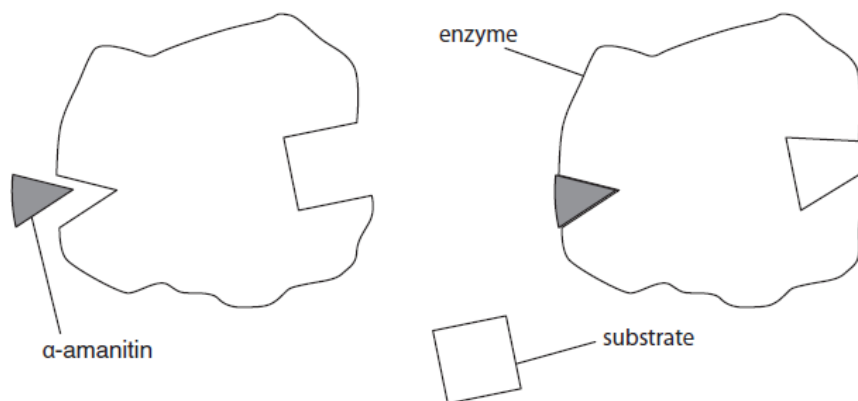


Fig. 2.2

- (i) Explain how α -amanitin stops the formation of an enzyme-substrate complex during transcription. [2]

2. (e) (ii) The Roman Emperor Claudius was poisoned by his wife Agrippina when she gave him death cap fungus to eat. The death cap fungus contains α -amanitin.

Suggest how the toxin α -amanitin may lead to the death of an organism.

[2]

[Total: 12]

3. In Lake Tanganyika in Africa, there are six species of fish of the genus *Tropheus* and a much larger number of distinctly coloured subspecies of each of the six species. *Tropheus* species are small fish that are confined to isolated rocky habitats around the shores of Lake Tanganyika.

The six species evolved during the primary radiation phase when the lake was first filled, about 1.25 million years ago. They arose from river dwelling ancestors and then filled all available niches in the lake.

Secondary radiations into the many subspecies occurred during the last 200 000 years. Sometime during this period, the water level in the lake fell, resulting in the formation of three separate lake basins. These basins persisted for many thousands of years before the water level rose again.

Fig. 3.1 shows an outline map of the lake and the location of the three temporary basins caused by lowering of lake levels.

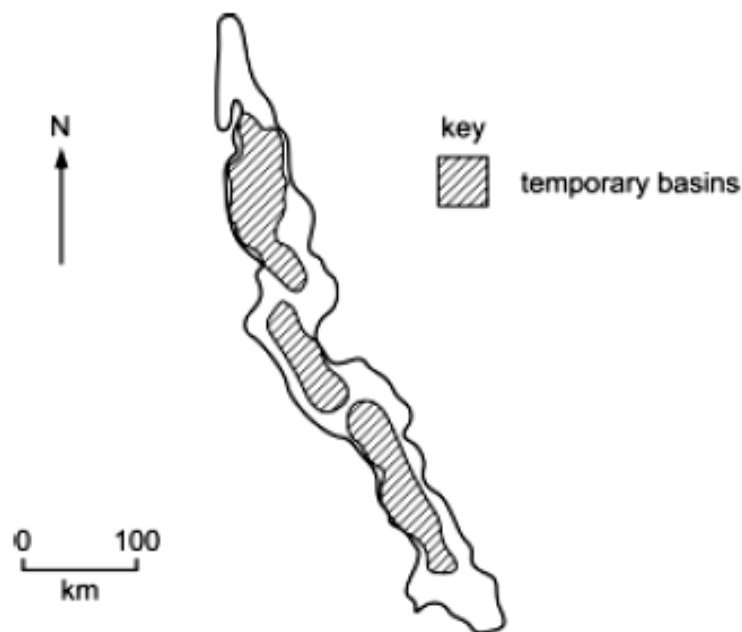


Fig. 3.1

- (a) Explain how natural selection could have caused the evolution of the six closely related species in the primary radiation.

[3]

Recent research has compared DNA sequence of cytochrome b gene of the six different species of *Tropheus* (**Fig. 3.2**).

<i>T. brichardi</i>	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
<i>T. kasabae</i>	CTTTTAACTATGATAATAGCTTTTGTGGGCTACG
<i>T. annectens</i>	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
<i>T. moorii</i>	CTTTTAACTATGATAACAGCTTTTGTGGGTTACG
<i>T. polli</i>	CTTTTAACTATAATAACAGCTTTTGTGGGCTACG
<i>T. duboisi</i>	CTTTTAACCATGATAACGGCTTTTGTAGGCTATG
	* *

Fig. 3.2 shows part of the cytochrome b DNA sequence

(b) With reference to **Fig. 3.2**,

- (i)** explain why different species of *Tropheus* can have similar DNA sequence in their cytochrome b gene. [2]

- (ii)** explain the advantage of the use of DNA sequence over amino acid sequence in determining the evolutionary relationship of *Tropheus* species. [2]

[Total: 7]

4.

PART I

Many bacteria make enzymes which protect the bacteria by breaking down the DNA molecules carried into the cell by viruses. The DNA target sites of four restriction enzymes are shown in **Table 4.1**. The points at which the strands of DNA are cut are shown by arrows and lines.

Table 4.1

restriction enzyme	target site
Sau3AI	$\begin{array}{c} \downarrow \text{G} - \text{A} - \text{T} - \text{C} - \\ - \text{C} - \text{T} - \text{A} - \text{G} \uparrow \end{array}$
BamHI	$\begin{array}{c} - \text{G} \downarrow \text{G} - \text{A} - \text{T} - \text{C} - \text{C} - \\ - \text{C} - \text{C} - \text{T} - \text{A} - \text{G} \uparrow \text{G} - \end{array}$
HinfI	$\begin{array}{c} - \text{G} \downarrow \text{A} - \text{N} - \text{T} - \text{C} - \\ - \text{C} - \text{T} - \text{N} - \text{A} \uparrow \text{G} - \end{array}$ 'N/N' may be any complementary base pair

(a) Suggest why these enzymes do not cut bacterial DNA.

[1]

(b) With reference to the information in **Table 4.1**,

(i) explain whether or not a piece of DNA cut by **Sau3AI** could join with one cut by **BamHI**;

[2]

(ii) show on **Fig. 4.1** the result of exposing this piece of DNA to **HinfI**.

$$\begin{array}{c} - \text{G} - \text{A} - \text{T} - \text{T} - \text{C} - \text{A} - \text{G} - \text{A} - \text{A} - \text{T} - \text{T} - \text{T} - \text{C} - \text{G} - \text{A} - \text{A} - \text{T} - \text{C} \\ - \text{C} - \text{T} - \text{A} - \text{A} - \text{G} - \text{T} - \text{C} - \text{T} - \text{T} - \text{A} - \text{A} - \text{A} - \text{G} - \text{C} - \text{T} - \text{T} - \text{A} - \text{G} \end{array}$$

Fig. 4.1

[1]

PART II

Fig. 4.2 shows a vector map of the plasmid vector pSING 003, which is used to clone the gene coding for human insulin.

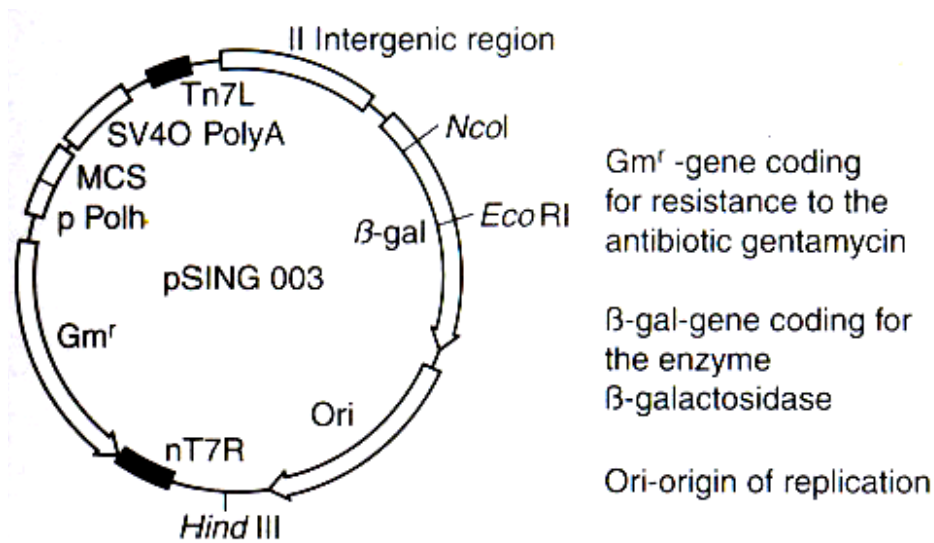


Fig. 4.2

- (c) Describe, with reference to specific restriction enzymes, how the insulin gene can be inserted into the vector pSING 003.

[3]

- (d) **Fig. 4.3** shows the results of plating the transformed cells onto an agar plate with the appropriate substances.

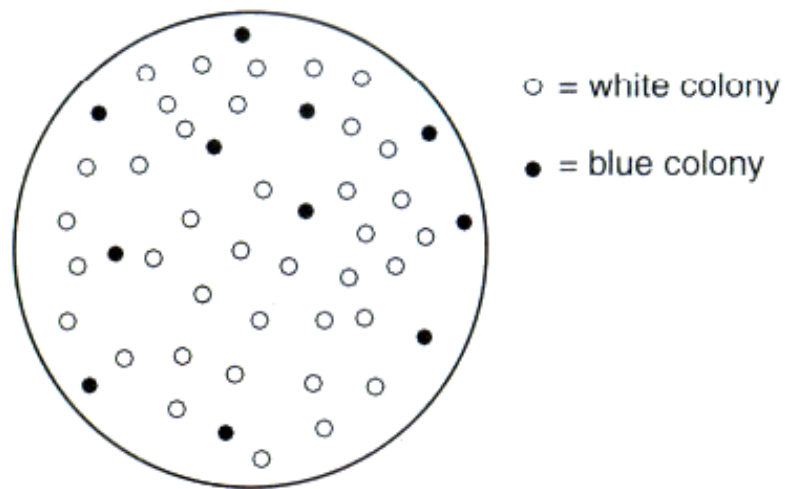


Fig. 4.3

- (i) Explain why some colonies appeared white while others appeared blue. [2]

- (ii) Suggest why replica plating was not necessary in this experiment. [2]

[Total: 11]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

Either

5. (a) Explain how the structure and properties of triglycerides are related to their functions in living organisms. [6]
- (b) Explain the role of membranes in photosynthesis and respiration. [8]
- (c) Compare haemoglobin with insulin in term of structure and function. [6]

[Total: 20]

OR

6. (a) Describe the unique features of stem cells (zygote, embryonic and adult stem cells) and explain their normal functions in a living organism. [7]
- (b) Describe the behaviour of DNA during the nuclear division process which takes place in the stem cells. [8]
- (c) Explain the effects of pH and temperature on the rate of enzyme-catalysed reactions. [5]

[Total: 20]